

BENCHMARK: INACCURATE

Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2461025.83885 +/- 0.0009 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.133 +/- 0.011                  |
| Transit depth (Rp/Rs)^2                           | 1.77 +/- 0.29 [%]                |
| Orbital Inclination (inc)                         | 83.36 +/- 1.62                   |
| Airmass coefficient 1 (a1)                        | 1.0777 +/- 0.0076                |
| Airmass coefficient 2 (a2)                        | -0.068 +/- 0.035                 |
| Scatter in the residuals of the lightcurve fit is | 0.7 %                            |
| Best Comparison Star                              | #2 - [374, 326]                  |
| Optimal Aperture                                  | 4.42                             |
| Optimal Annulus                                   | 10.1                             |
| Transit Duration (day)                            | 0.1261 +/- 0.0035                |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.133 ± 0.011 vs 0.1178 (deviation 1.38σ) |
| Duration fit vs published | 0.1261 d vs 0.125 d (deviation 0.31σ)     |
| Mid-time O–C              | -4.4 min                                  |
| Depth SNR                 | 12.1                                      |
| Residual scatter          | 0.7 %                                     |

Light curve

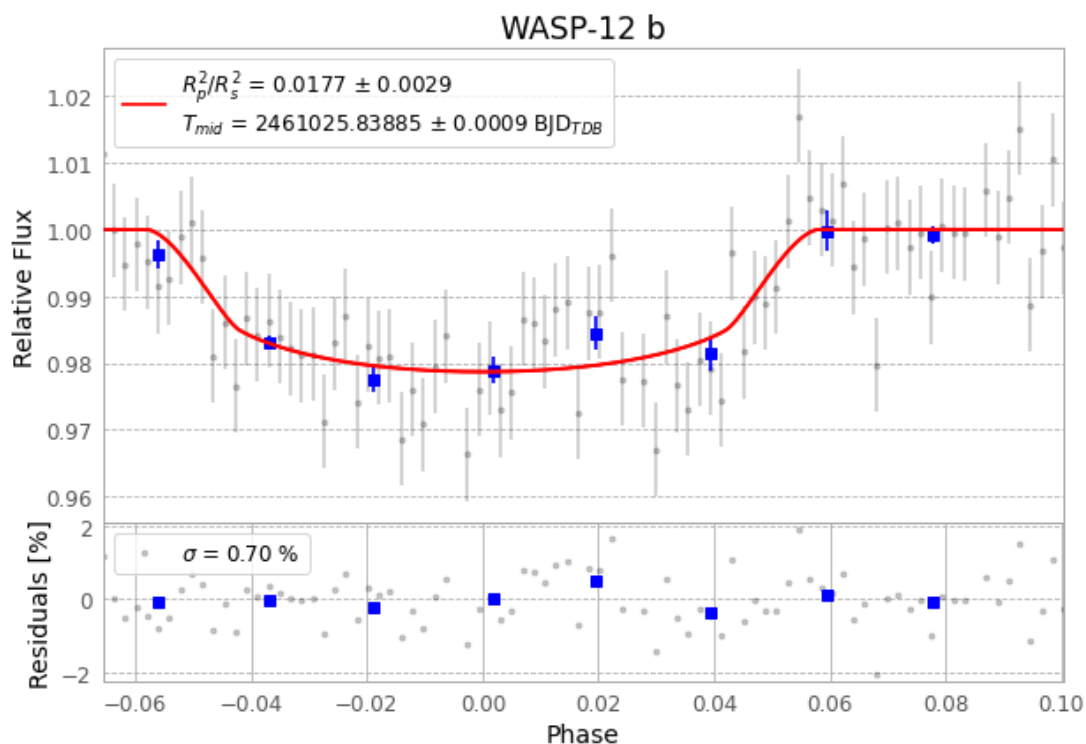


Figure 1 — FinalLightCurve\_WASP-12 b\_2025-12-15.png (SHA-256 abb55a98529032d6...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 295], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 c835b7dcc8840c0c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

## Data provenance

87 FITS frames (57 MB), WASP-12251216061516.FITS → WASP-12251216103616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-251216.log (102aa7adaca8...), FinalLightCurve\_WASP-12 b\_2025-12-15.png (abb55a985290...), FinalLightCurve\_WASP-12 b\_2025-12-15.csv (22eeeb10f84d...)

## Compute resources

Wall time 170.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:59Z.